Capstone Project Report

Submitted By

Abhishek Shinde

Organization

Bluestock Fintech

Project Duration

01 June 2026 – 12 June 2026

Executive Summary

The Mutual Fund Analytics Platform was developed to analyze mutual fund industry trends, fund performance, investor behavior, and portfolio allocations using data analytics techniques. The project involved building an end-to-end data pipeline, performing exploratory data analysis, calculating performance metrics, implementing advanced risk analytics, and creating an interactive dashboard.

The system provides insights into Assets Under Management (AUM), SIP inflows, NAV trends, investor demographics, fund performance, portfolio concentration, and risk measures such as Sharpe Ratio, VaR, CVaR, and Maximum Drawdown. The final solution helps investors and analysts understand mutual fund performance and market behavior through data-driven decision-making.

Project Objectives

The primary objectives of the project were:

1. Collect and process mutual fund datasets.
2. Build an ETL pipeline for data cleaning and transformation.
3. Perform exploratory data analysis on industry trends.
4. Analyze mutual fund performance using financial metrics.
5. Build an interactive dashboard for business users.
6. Implement advanced risk analytics and investor behavior analysis.
7. Generate reports and recommendations from analytical findings.

Data Sources

The project utilized the following datasets:

1. Fund Master Data
2. NAV History Data
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count Data
7. Scheme Performance Data
8. Investor Transactions Data
9. Portfolio Holdings Data
10. Benchmark Index Data

The datasets were provided in CSV format and processed using Python and Pandas.

Technology Stack

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- SciPy

Database:

- SQLite

Visualization:

- Power BI

Version Control:

- Git & GitHub

ETL Pipeline Design

The ETL pipeline was designed to transform raw mutual fund datasets into cleaned analytical datasets.

Steps:

1. Data Ingestion
    - Load raw CSV datasets.
2. Data Cleaning
    - Remove duplicates.
    - Handle missing values.
    - Standardize formats.
3. Data Transformation
    - Convert date columns.
    - Create derived metrics.
    - Aggregate data where required.

4. Data Storage
    - Save processed datasets.
    - Store data in SQLite database.
5. Analytics Layer
    - EDA Analysis
    - Performance Analytics
    - Advanced Analytics

Exploratory Data Analysis (EDA)

Several visualizations were created to understand industry trends.

NAV Trend Analysis

Daily NAV values were analyzed across multiple schemes from 2022 to 2026. The analysis highlighted strong growth during bullish market phases and temporary declines during correction periods.

AUM Growth Analysis

AUM trends across fund houses indicated significant growth over the analysis period. SBI Mutual Fund emerged as one of the largest contributors in terms of total assets under management.

SIP Inflow Analysis

Monthly SIP inflows showed a consistent upward trend, indicating increasing retail investor participation in mutual funds.

Category Inflow Analysis

Different fund categories attracted varying levels of investor capital. Equity-oriented categories generally experienced stronger inflows compared to debt-oriented categories.

Investor Demographics

Analysis of investor demographics revealed variations in investment behavior across age groups, genders, and geographic locations.

Geographic Distribution

State-wise analysis highlighted the concentration of investments in major financial centers while also showing growth in emerging investment regions.

Fund Performance Analytics

Several performance metrics were calculated for evaluating mutual funds.

CAGR

Compound Annual Growth Rate (CAGR) was calculated over multiple investment horizons to compare long-term fund performance.

Sharpe Ratio

Sharpe Ratio was used to evaluate risk-adjusted returns. Funds with higher Sharpe Ratios delivered better returns relative to the risks undertaken.

Sortino Ratio

Sortino Ratio focused specifically on downside risk and provided additional insight into fund stability.

Alpha

Alpha measured the excess return generated by funds relative to benchmark indices.

Beta

Beta quantified fund sensitivity to market movements.

Maximum Drawdown

Maximum Drawdown measured the largest peak-to-trough decline experienced by a fund during the analysis period.

Fund Scorecard

A composite scorecard was developed using multiple performance indicators to rank mutual funds objectively.

Advanced Analytics and Risk Metrics

Advanced analytical techniques were implemented to enhance investment insights.

Value at Risk (VaR)

Historical VaR at the 95% confidence level was computed to estimate potential losses under adverse market conditions.

Conditional Value at Risk (CVaR)

CVaR measured expected losses beyond the VaR threshold and provided deeper downside risk assessment.

Rolling Sharpe Ratio

Rolling 90-day Sharpe Ratios were calculated to evaluate changes in risk-adjusted performance over time.

Investor Cohort Analysis

Investors were grouped based on their first transaction year. Cohort analysis helped identify long-term participation patterns.

SIP Continuity Analysis

Investor transaction patterns were examined to identify SIP interruptions and potential at-risk investors.

Fund Recommendation Engine

A simple recommendation engine was developed to suggest funds based on investor risk appetite categories such as Low, Moderate, and High.

Portfolio Concentration Analysis

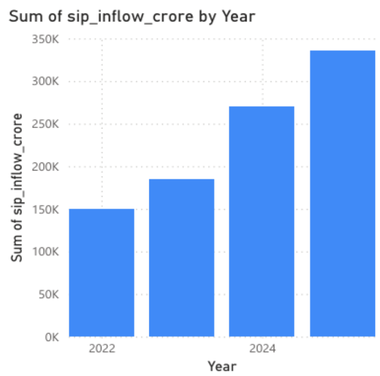
Herfindahl-Hirschman Index (HHI) was calculated to evaluate portfolio diversification levels across mutual funds.

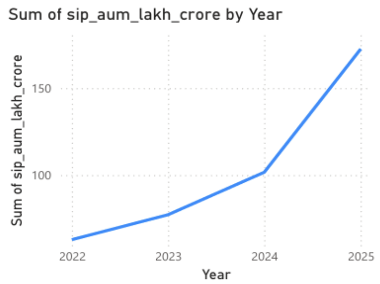
Dashboard Development

An interactive Power BI dashboard was developed with four major sections:

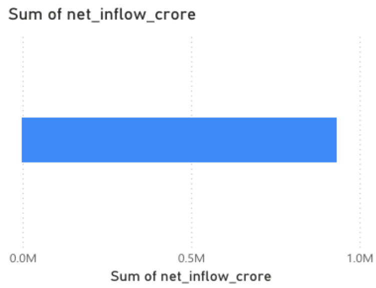
1. Industry Overview
2. Fund Performance
3. Investor Analytics
4. SIP and Market Trends

The dashboard enables users to interact with data through filters and visualizations.





category	2024	2025	Total
ELSS	4627	1453	**6080**
Flexi Cap	47551	16438	**63989**
Gilt	7753	2642	**10395**
Hybrid	29711	9157	**38868**
Large & Mid Cap	43169	14583	**57752**
Large Cap	19449	6184	**25633**
Liquid	346328	104947	**451275**
Mid Cap	41116	14196	**55312**
Sectoral/Thematic	78107	25722	**103829**
Short Duration	41859	13671	**55530**
Small Cap	34204	12392	**46596**
Value/Contra	12744	4236	**16980**
Total	**706618**	**225621**	**932239**



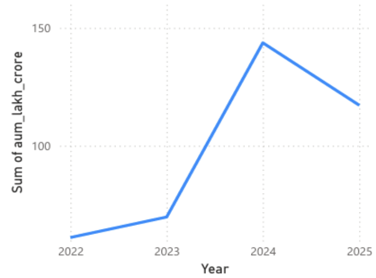
12.50
Max of aum_lakh_crore

31K
SIP Inflows

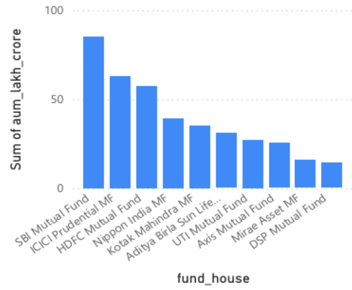
26.12
Folios

ABSL Frontlin...
Schemes

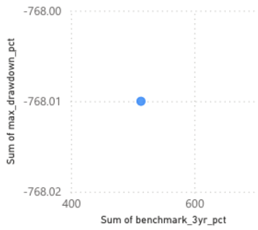
Sum of aum_lakh_crore by Year



Sum of aum_lakh_crore by fund_house



Sum of benchmark_3yr_pct and Sum of max_drawdown_pct

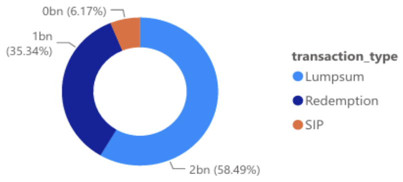


fund_house

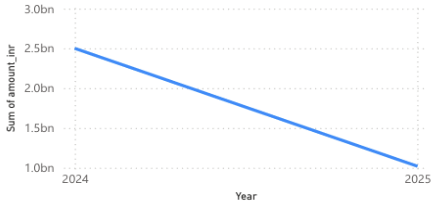
Adity...	Axis ...	DSP ...
HDFC...	ICICI ...	Kotak...
Mirae...	Nipp...	SBI M...

amfi_code	fund_house	Sum of benchmark_3yr_pct	Sum of alpha	Sum of beta	Sum of max_drawdown_pct
100016	HDFC Mutual Fund	14.06	0.78	0.97	-17.41
100025	HDFC Mutual Fund	5.39	1.98	0.44	-6.01
100033	HDFC Mutual Fund	15.63	0.95	0.91	-13.67
101206	Aditya Birla Sun Life MF	12.44	1.34	1.03	-15.07
101207	Aditya Birla Sun Life MF	20.54	1.84	0.97	-23.61
101208	Aditya Birla Sun Life MF	3.96	1.18	0.43	-3.66
102885	UTI Mutual Fund	11.17	0.93	0.90	-24.42
102886	UTI Mutual Fund	14.49	1.12	0.92	-13.43
102887	UTI Mutual Fund	13.55	1.79	1.00	-12.14
118632	Nippon India MF	13.14	0.86	0.88	-16.07
118633	Nippon India MF	10.63	1.70	1.02	-18.14
118634	Nippon India MF	19.35	0.80	1.03	-30.87
118635	Nippon India MF	9.97	1.80	1.04	-26.75
118636	Nippon India MF	4.42	0.89	0.37	-2.23
119092	Axis Mutual Fund	10.43	1.41	0.91	-27.54
119093	Axis Mutual Fund	10.71	1.43	0.87	-17.40
119094	Axis Mutual Fund	13.76	1.42	1.00	-32.38
119095	Axis Mutual Fund	20.47	0.51	1.00	-14.45
119120	SBI Mutual Fund	4.47	1.60	0.22	-2.30
119551	SBI Mutual Fund	11.49	0.87	0.89	-21.70
119552	SBI Mutual Fund	9.52	1.78	0.87	-24.43
119598	SBI Mutual Fund	22.16	1.23	0.89	-13.35
119599	SBI Mutual Fund	22.01	1.13	1.04	-24.78
120503	ICICI Prudential MF	10.88	0.66	0.96	-25.91
120504	ICICI Prudential MF	13.53	0.88	1.03	-26.59
120505	ICICI Prudential MF	17.19	0.89	1.00	-21.84
120506	ICICI Prudential MF	14.21	0.55	0.92	-21.89
120507	ICICI Prudential MF	5.83	1.85	0.26	-2.62
120841	Kotak Mahindra MF	10.98	1.27	0.93	-17.86
Total		**513.42**	**50.14**	**34.93**	**-768.01**

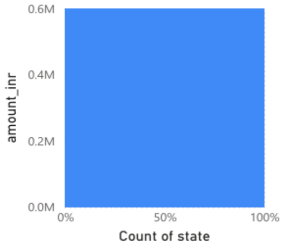
Sum of amount_inr by transaction_type



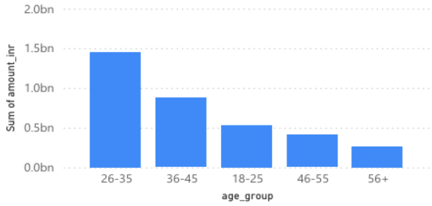
Sum of amount_inr by Year



Count of state by amount_inr



Sum of amount_inr by age_group



Key Findings

1. Mutual fund industry AUM increased significantly from 2022 to 2025.
2. SIP inflows reached record levels during the analysis period.
3. Equity-oriented funds attracted higher investor participation.
4. High-return funds generally exhibited higher risk measures.
5. Investor participation continued to increase across all age groups.
6. Several funds demonstrated strong risk-adjusted performance.
7. Portfolio concentration varied considerably across schemes.
8. Investor cohorts showed increasing long-term engagement.
9. Rolling Sharpe Ratios fluctuated during periods of market volatility.
10. Risk analytics provided valuable insights into downside exposure.

Limitations

1. Analysis is based on historical data.
2. Real-time market data integration was not implemented.
3. External economic factors were not explicitly modeled.
4. Some assumptions were made while calculating risk metrics.
5. Dashboard functionality depends on the availability of source datasets.

Recommendations

1. Integrate real-time market feeds.
2. Enhance recommendation models using machine learning.
3. Add automated report generation.
4. Deploy dashboard to a cloud platform.
5. Implement portfolio optimization techniques.
6. Incorporate macroeconomic indicators into risk models.

Conclusion

The Mutual Fund Analytics Platform successfully demonstrated the application of data engineering, exploratory analysis, performance analytics, risk modeling, and dashboard development within the mutual fund domain.

The project provided meaningful insights into mutual fund performance, investor behavior, and market trends while establishing a strong analytical foundation for future enhancements.

References

1. AMFI India
2. NSE India
3. SEBI Mutual Fund Guidelines
4. Pandas Documentation
5. NumPy Documentation
6. Power BI Documentation
7. SQLite Documentation
8. Python Documentation
